

Week 1 Module – C & C++ Programming

Topics to be Covered

Students are required to Submit handwritten notes and practice for the following topics in **C and C++ Programming Language**:

1. **Syntax**
2. **Conditions and Loops**
3. **Programming Problems**
 - Solve the given **Star Pattern Questions**

1. Square

```
*****
*****
*****
*****
*****
```

2. Hollow Square

```
*****
*   *
*   *
*   *
*   *
*****
```

3. Triangle

```
*
**
***
****
*****
```

4. Inverted Triangle

```
*****
****
***
**
*
```

5. Pyramid

```
  *
 ***
*****
*****
*****
```

6. Inverted Pyramid

```
*****
*****
*****
***
*
```

7. Half Diamond

```
*
**
***
****
*****
****
***
**
*
```

8. Diamond

```
  *
 ***
*****
*****
*****
*****
***
 *
*
```

Submission Guidelines

Students must prepare a **single PDF document** containing the following:

1. Handwritten Notes

- Handwritten notes for **each topic/module**
- Proper explanation of concepts in a notebook

2. Handwritten Code

- Write all program codes in the notebook
- Include proper formatting and comments (if applicable)

3. Executable Code Proof

- Run all programs on your system
- Attach **screenshots of executable/output code** for every program in the PDF

PDF Requirements

The final PDF must include:

- ✓ Handwritten theory notes
- ✓ Handwritten code in notebook
- ✓ Screenshots of successfully executed programs/output

GitHub Submission

- Push the **PDF file** to a GitHub repository.
- Repository name format must be: **2nd_Yr_Week1_Module** (Use the exact naming format.)

Important Instructions

- Submit work neatly and in proper sequence.
- Make sure screenshots are clearly visible.
- Incomplete submissions (without handwritten notes/code or execution screenshots) will not be considered.